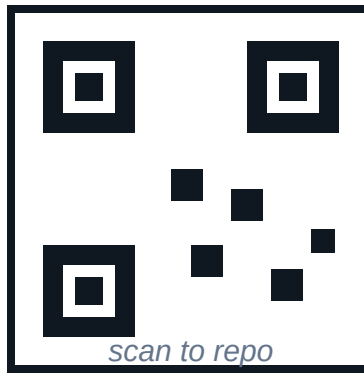


Evidence-Centered Visualization for ML Security Engineering



Aggregating Repository-Level Measurements Into Traceable Technical Dashboards

Pooja Kiran

Security Engineering • AI / Agent Security • Cloud Security

MIT • Python 3.12 • HEAD 89a038a • verified 2026-09-26

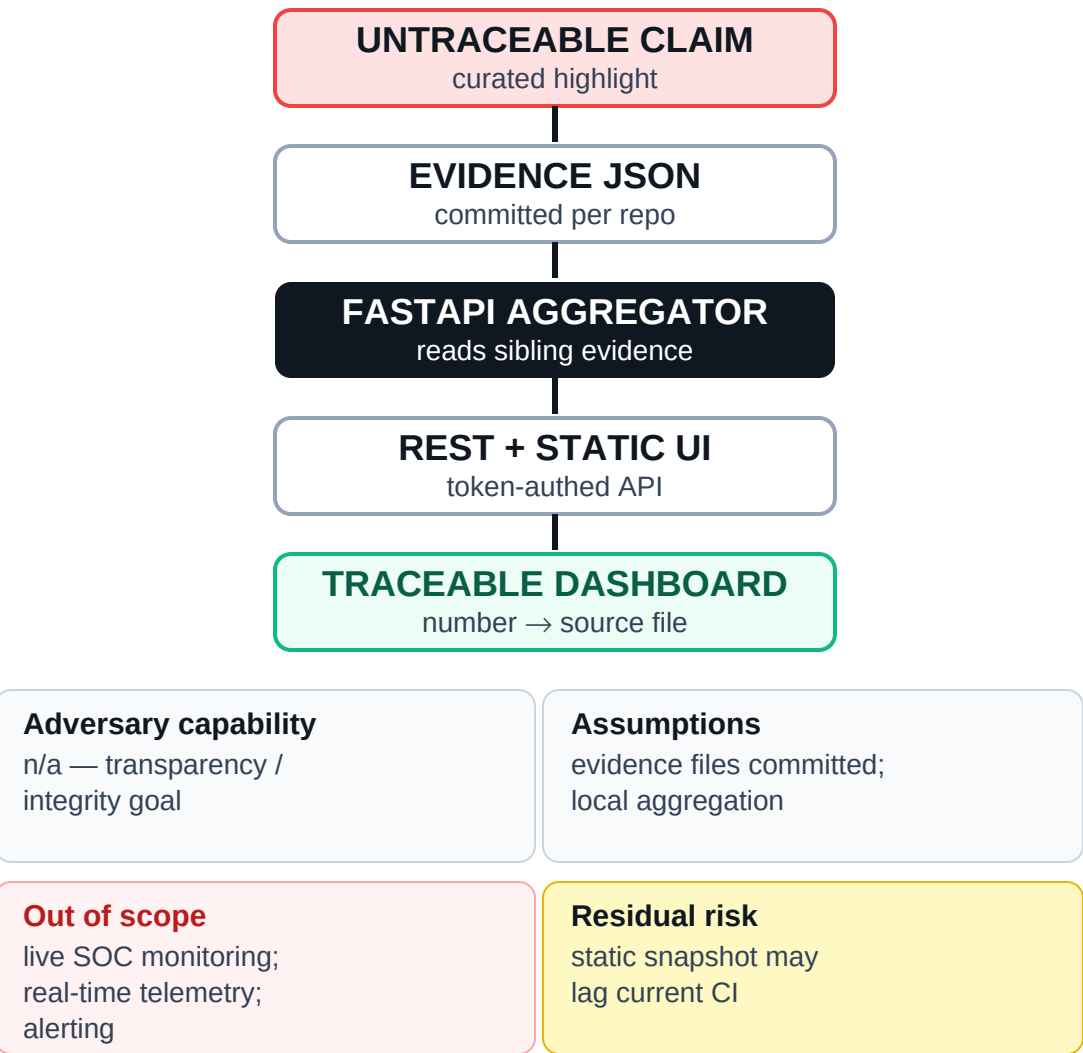
RESEARCH QUESTION

Can ML security claims be presented so each number traces back to a committed evidence file — including the weak results, not just the strong?

01 • THE FAILURE MODE

Security dashboards often show curated highlights with no path back to the underlying measurement. A viewer cannot tell a real number from a hopeful one. This server reads committed JSON evidence from sibling repos and renders dashboards that show weak results next to strong ones, each traceable to its source file.

02 • THREAT MODEL



03 • OBJECTIVE

Determine whether committed per-repo evidence can be aggregated into honest dashboards where every metric traces to its source artifact.

- 01 – Discover sibling repo evidence
- 02 – Per-project HTML dashboards
- 03 – Token-authed aggregated REST API
- 04 – Show weak results explicitly

HERO • EVIDENCE AGGREGATION

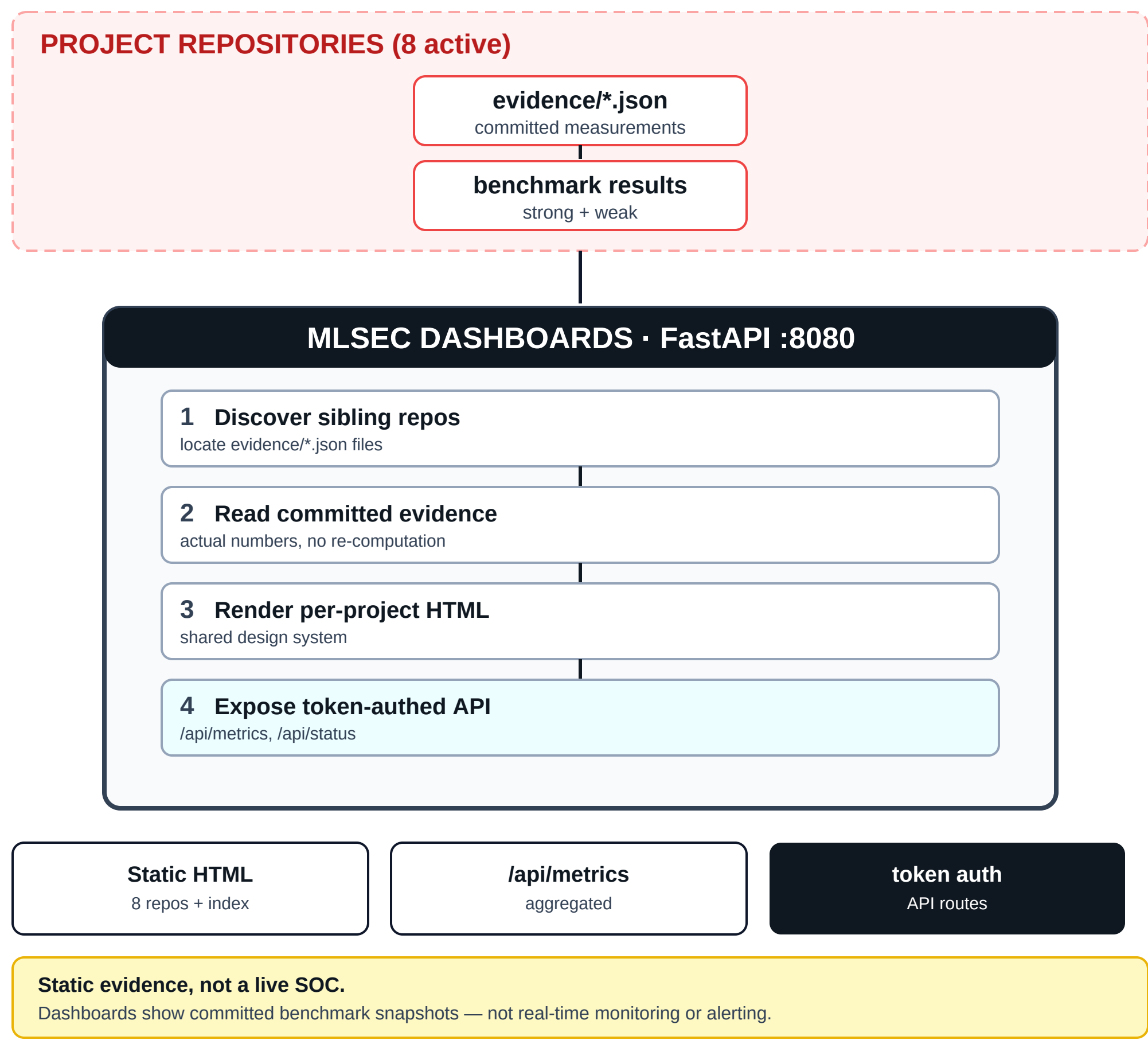
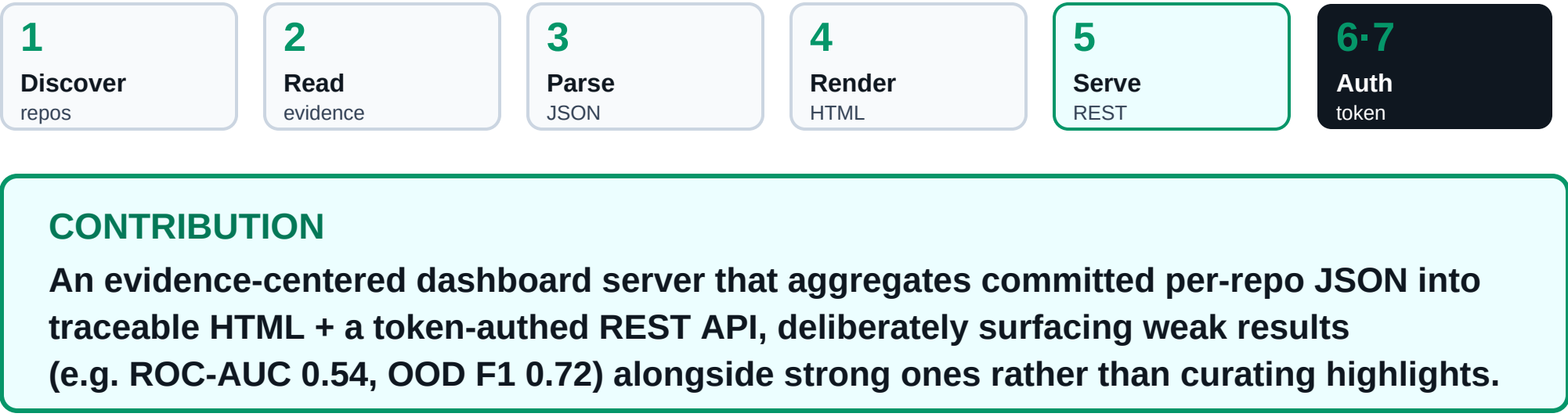
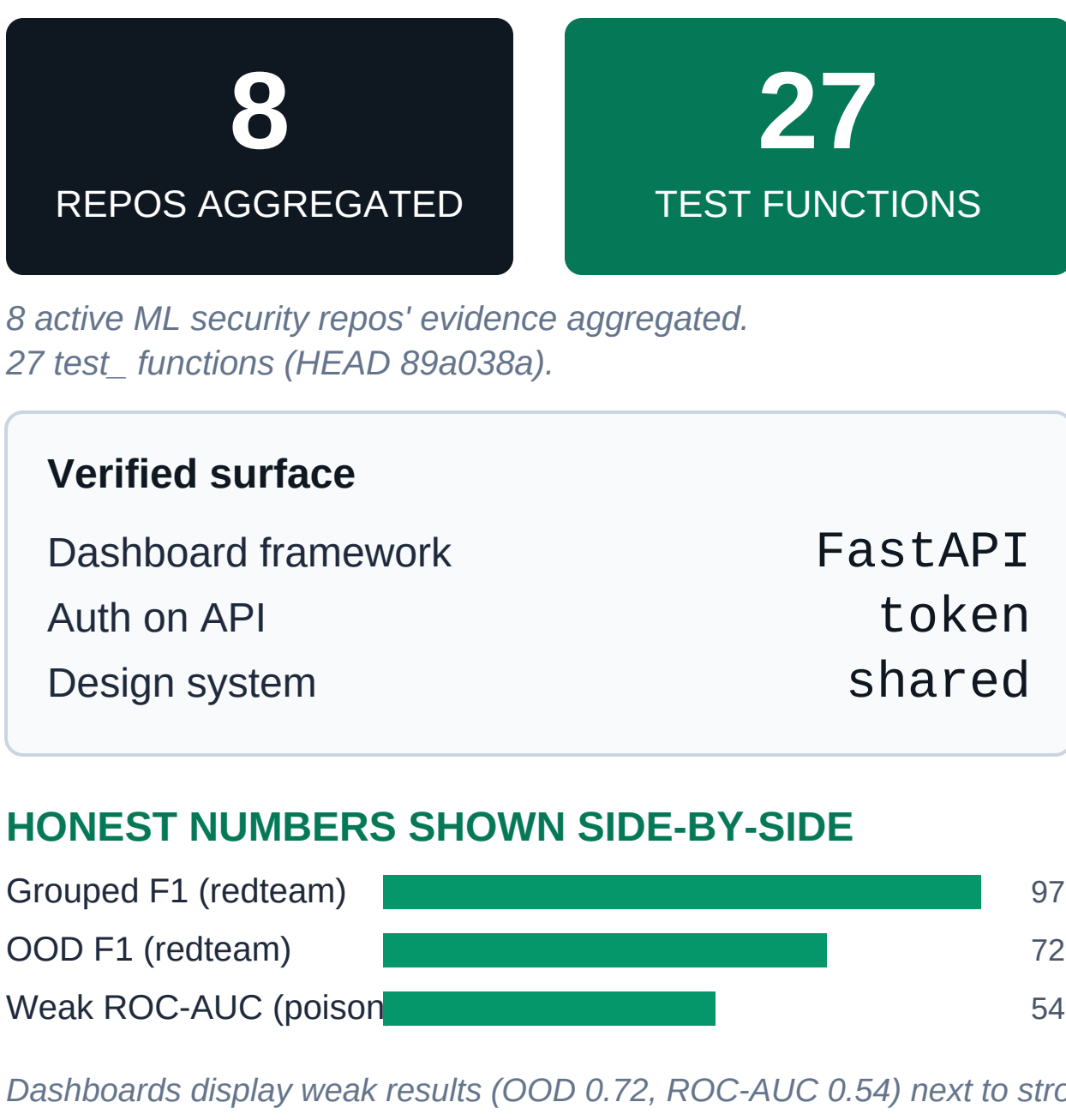


Figure 1. A FastAPI aggregator reads committed evidence JSON from sibling repos and renders traceable dashboards — weak results shown too.

METHOD • AGGREGATION FLOW



RESULTS • CURRENT LOCAL VALIDATION



Dashboards display weak results (OOD 0.72, ROC-AUC 0.54) next to strong ones — by design.

TRANSPARENCY MECHANISM

Every displayed number is read from a committed evidence/*.json in a sibling repo — not hand-entered. Weak results are shown, not hidden.

NOT ESTABLISHED BY THIS REPO

Live SOC monitoring. Real-time telemetry or alerting. That static snapshots equal current CI state.

SECURITY BOUNDARY

PROTECTS / DETECTS	DOES NOT ESTABLISH
- evidence traceability - weak-result honesty - token-authed API - per-repo dashboards - shared design	- live monitoring - real-time alerts - SOC operation - auto data refresh - threat response

CLAIMS → EVIDENCE

LIMITATIONS

- 1. Static evidence snapshots, not live monitoring.
- 2. Data can lag the current CI state of each repo.
- 3. No alerting or real-time telemetry.
- 4. Depends on sibling repos committing evidence.
- 5. Local aggregation; not a hardened service.

FUTURE WORK

- Scheduled evidence refresh from CI.
- Historical trend visualization.
- Signed-evidence verification on ingest.
- Diff view across commits.
- Per-metric provenance drill-down.

CONCLUSION

The server aggregates committed per-repo evidence into traceable dashboards and a token-authed API, showing weak results beside strong ones. It makes claims auditable — it is not a live SOC and does not perform real-time monitoring or alerting.

REPRODUCE

```
python dashboard_server.py
pytest tests/
```